

Chapter 4

Project Integration Management

The Key to Overall Project Success: Good Project Integration Management

- Project managers must coordinate all of the other knowledge areas throughout a project's life cycle
- Project integration management is *not* the same thing as software integration

Project Integration Management Processes

Process Group	Integration Management Process	Major Output
Initiating	Developing the project charter	Project Charter
Planning	Developing the project management plan	Project Management Plan
Executing	Directing and managing project work	Deliverables, work performance info, change requests
Monitoring and Controlling	(1) Monitor/Control Project Work (2) Perform Integrated Change Control	(1) Change Requests (2) Status updates
Closing	Close Project/Phase	Product Transition

Project Integration Management Summary

Initiating

Process: **Develop project charter**

Output: Project charter

Planning

Process: **Develop project management plan**

Output: Project management plan

Executing

Process: **Direct and manage project work**

Outputs: Deliverables, work performance data, change requests, project management plan updates, project documents updates

Monitoring and Controlling

Process: **Monitor and control project work**

Outputs: Change requests, project management plan updates, project documents updates

Process: **Perform integrated change control**

Outputs: Approved change requests, change log, project management plan updates, project documents updates

Closing

Process: **Close project or phase**

Outputs: Final product, service, or result transition; organizational process assets updates

Project Start

Project Finish

Strategic Planning and Project Selection

- **Strategic planning** involves determining long-term objectives, predicting future trends, and projecting the need for new products and services
- Organizations often perform a **SWOT analysis**

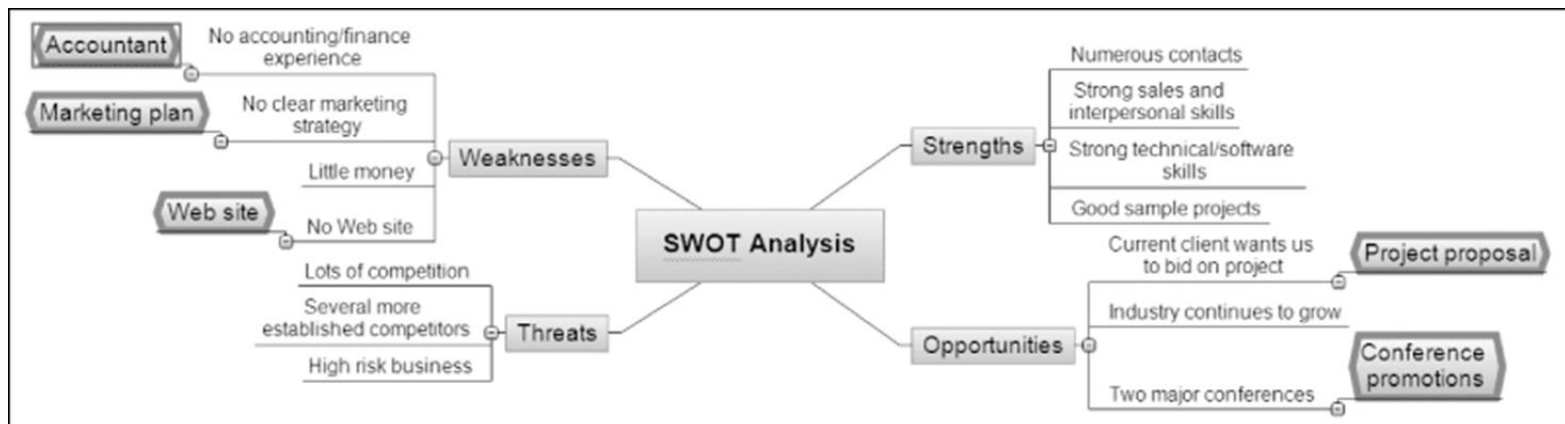
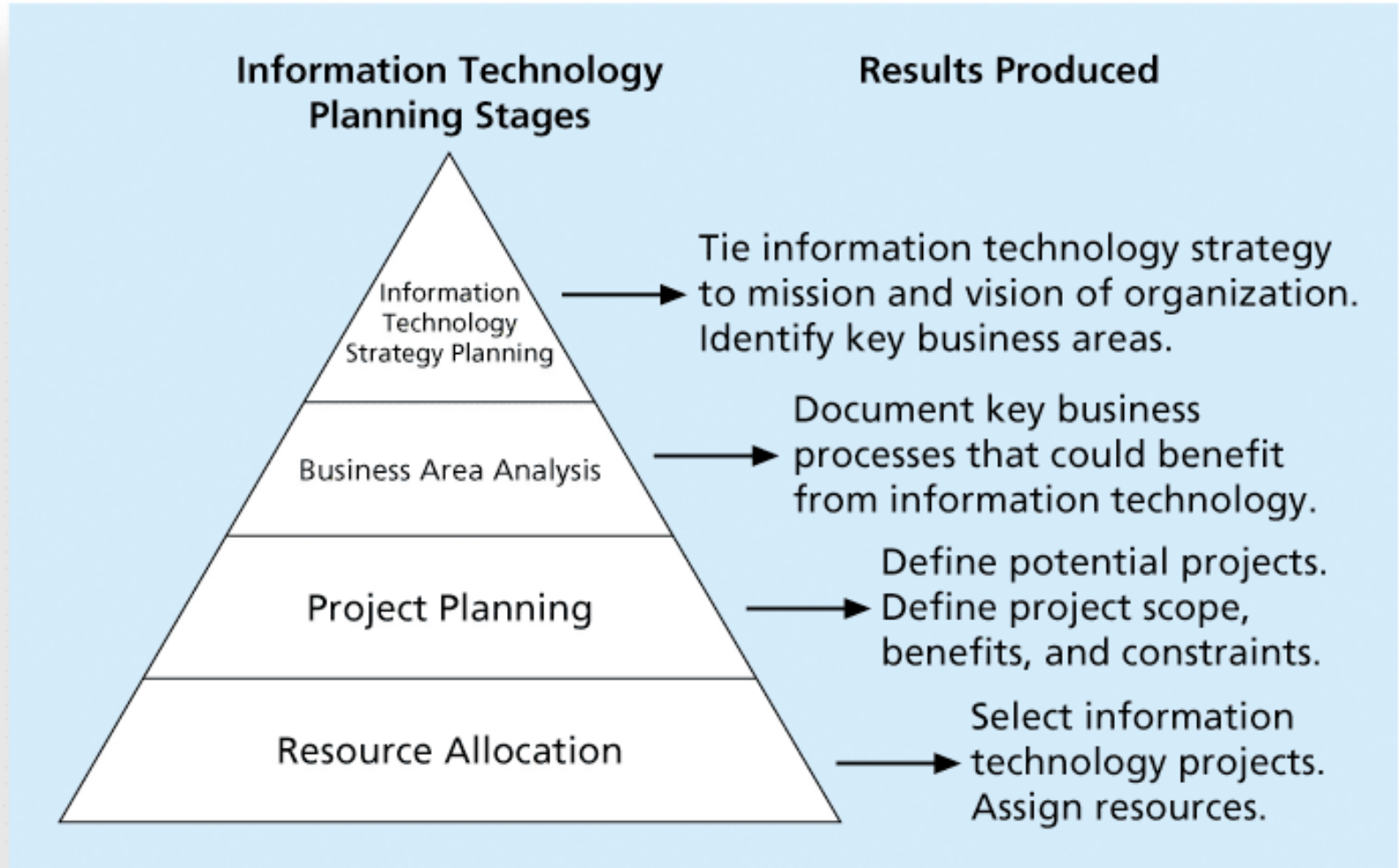


Figure 4-3. Information Technology Planning Process



Learning from Project Success

- Only one in seven product concepts comes to fruition. Why is it that some companies, like Proctor & Gamble, Johnson and Johnson, Hewlett Packard, and Sony are consistently successful in NPD? Because they use a disciplined, systematic approach to NPD projects based on best practices
- Four important forces behind NPD success include the following:
 1. A product innovation and technology strategy for the business
 2. Resource commitment and focusing on the right projects, or solid portfolio management
 3. An effective, flexible and streamlined idea-to-launch process
 4. The right climate and culture for innovation, true cross-functional teams, and senior management commitment to NPD

Methods for Selecting Projects

- There are usually more projects than available time and resources to implement them

- **Methods for selecting projects**
 - Feasibility Study – can it even be done?
 - Benefits Contributions
 - Economics Models
 - Scoring Models

Method: Focusing on Broad Organizational Needs

- It is often difficult to provide strong justification for many IT projects, but everyone agrees they have a high value
- Three important criteria for projects

Method: Categorizing IT Projects

- One categorization is whether the project addresses
 - a problem
 - an opportunity, or
 - a directive

Methods: Financial Analysis of Projects

- Three primary methods for determining the projected financial value of projects:
 - Net present value (NPV) analysis
 - Return on investment (ROI)
 - Payback analysis

Net Present Value Analysis

- **Net present value (NPV)** analysis is a method of calculating the expected net monetary gain or loss from a project by discounting all expected future cash inflows and outflows to the present point in time

	A	B	C	D	E	F	G
1	Discount rate	10%					
2							
3	PROJECT 1	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
4	Benefits	\$0	\$2,000	\$3,000	\$4,000	\$5,000	\$14,000
5	Costs	\$5,000	\$1,000	\$1,000	\$1,000	\$1,000	\$9,000
6	Cash flow	(\$5,000)	\$1,000	\$2,000	\$3,000	\$4,000	\$5,000
7	NPV →	\$2,316					
8		Formula =npv(b1,b6:f6)					
9							
10	PROJECT 2	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
11	Benefits	\$1,000	\$2,000	\$4,000	\$4,000	\$4,000	\$15,000
12	Costs	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$10,000
13	Cash flow	(\$1,000)	\$0	\$2,000	\$2,000	\$2,000	\$5,000
14	NPV →	\$3,201					
15		Formula =npv(b1,b13:f13)					
16							
17							

Note that totals are equal, but NPVs are not because of the time value of money

NPV Calculations

- Determine estimated costs and benefits for the life of the project and the products it produces
- Determine the discount rate (check with your organization on what to use)
- Calculate the NPV

$$NPV = \sum_{t=0...n} A_t / (1+r)^t$$

A_t = Amount of Cash Flow

t = year of the cash flow

r = discount rate

Return on Investment

- **Return on investment (ROI)** is calculated by subtracting the project costs from the benefits and then dividing by the costs

$$\text{ROI} = (\text{total discounted benefits} - \text{total discounted costs}) / \text{discounted costs}$$

JWD Consulting NPV Example

Discount Rate: 8%

Costs:

- Y0: \$140,000
- Y1 - 3: \$40,000

Benefits:

- Y0: \$0
- Y1 - 3: \$200,000

Year 0: discount factor = $1/(1+0.08)^0 = 1$

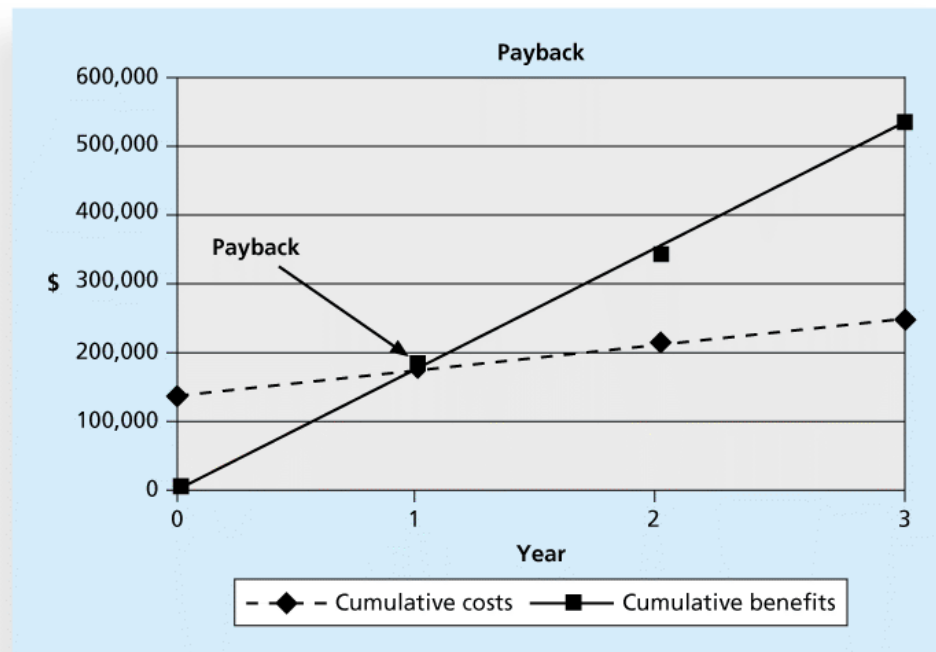
Year 1: discount factor = $1/(1+0.08)^1 = 0.93$

JWD Consulting NPV Example

Discount rate	8%					
Assume the project is completed in Year 0			Year			
	0	1	2	3	Total	
Costs	140,000	40,000	40,000	40,000		
Discount factor	1	0.93	0.86	0.79		
Discounted costs	140,000	37,200	34,400	31,600	243,200	
Benefits	0	200,000	200,000	200,000		
Discount factor	1	0.93	0.86	0.79		
Discounted benefits	0	186,000	172,000	158,000	516,000	
Discounted benefits - costs	(140,000)	148,800	137,600	126,400	272,800	← NPV
Cumulative benefits - costs	(140,000)	8,800	146,400	272,800		
ROI	112%					
	Payback In Year 1					

Payback Analysis

- The **payback period** is the amount of time it will take to recoup, in the form of net cash inflows, the total dollars invested in a project



Method: Weighted Scoring Model

- A weighted scoring model is a tool that provides a systematic process for selecting projects based on many criteria
 - Identify criteria important to the project selection process
 - Assign weights (percentages) to each criterion so they add up to 100%
 - Assign scores to each criterion for each project
 - Multiply the scores by the weights and get the total weighted scores

Sample Weighted Scoring Model Criteria

- Criteria:
 - Supports key business objectives (25%)
 - Strong internal sponsor (15%)
 - Strong customer support (10%)
 - One year or less implementation (20%)
 - Positive NPV (30%)

Sample Weighted Scoring Model - Projects

■ **Project 1**

A director has been charged with finding ways to address the strategic goals for technology set at the recent board meeting. He has suggested an upgrade to the system that will take about a year to complete. Because their system is currently quite complex, this is based on estimates which could change frequently. He has support from many executives to move forward and, if they stay on budget, there will be a positive NPV.

■ **Project 2**

A manager wants to create an internal site to track project success and failures. She estimates a 6 month timeline with minimal risk of running longer or costing more than estimates. For right now, the NPV is negative.

Sample Weighted Scoring Model - Projects

■ **Project 3**

New system implementation that may take 18 months to complete. This implementation is being supported by the CIO and executives as it will enable the company to meet the new strategy to become a leader in customer service. The payback period is 4 years but it will result in positive NPV.

Method: Implementing a Balanced Scorecard

- A **balanced scorecard** is a methodology that converts an organization's value drivers, such as customer service, innovation, operational efficiency, and financial performance, to a series of defined metrics



Additional Methods

- Scoring
- Opportunity Costs
- Six Sigma

Initiating Process Group: Developing a Project Charter

- A **project charter** is a document that formally recognizes the existence of a project and provides direction on the project's objectives and management

Key Output of Project Integration:
Signed Charter

Inputs for Developing a Project Charter

Project Title: DNA-Sequencing Instrument Completion Project

Date of Authorization: February 1

Project Start Date: February 1

Projected Finish Date: November 1

Key Schedule Milestones:

- Complete first version of the software by June 1
- Complete production version of the software by November 1

Budget Information: The firm has allocated \$1.5 million for this project, and more funds are available if needed. The majority of costs for this project will be internal labor. All hardware will be outsourced.

Project Manager: Nick Carson, (650) 949-0707, ncarson@dnaconsulting.com

Project Objectives: The DNA-sequencing instrument project has been underway for three years. It is a crucial project for our company. This is the first charter for the project, and the objective is to complete the first version of the software for the instrument in four months and a production version in nine months.

Main Project Success Criteria: The software must meet all written specifications, be thoroughly tested, and be completed on time. The CEO will formally approve the project with advice from other key stakeholders.

Table 4-1. Project Charter (cont.)

Approach:

- Hire a technical replacement for Nick Carson and a part-time assistant as soon as possible.
- Within one month, develop a clear work breakdown structure, scope statement, and Gantt chart detailing the work required to complete the DNA sequencing instrument.
- Purchase all required hardware upgrades within two months.
- Hold weekly progress review meetings with the core project team and the sponsor.
- Conduct thorough software testing per the approved test plans.

ROLES AND RESPONSIBILITIES

Name	Role	Position	Contact Information
Ahmed Abrams	Sponsor	CEO	aabrams@dnaconsulting.com
Nick Carson	Project Manager	Manager	ncarson@dnaconsulting.com
Susan Johnson	Team Member	DNA expert	sjohnson@dnaconsulting.com
Renyong Chi	Team Member	Testing expert	rchi@dnaconsulting.com
Erik Haus	Team Member	Programmer	ehaus@dnaconsulting.com
Bill Strom	Team Member	Programmer	bstrom@dnaconsulting.com
Maggie Elliot	Team Member	Programmer	melliot@dnaconsulting.com

Sign-off: (Signatures of all the above stakeholders)

Ahmed Abrams

Susan Johnson

Erik Haus

Maggie Elliot

Nick Carson

Renyong Chi

Bill Strom

Comments: (Handwritten or typed comments from above stakeholders, if applicable)

"I want to be heavily involved in this project. It is crucial to our company's success, and I expect everyone to help make it succeed." —Ahmed Abrams

"The software test plans are complete and well documented. If anyone has questions, do not hesitate to contact me." —Renyong Chi

Planning Process Group:

Develop Project Management Plan

- A **project management plan** is a document used to coordinate all project planning documents and help guide a project's execution and control

- Common Elements:
 - Introduction or overview of the project
 - Description of how the project is organized
 - Management and technical processes
 - Work to be done, schedule, and budget info

Table 4-2. Sample Contents for a Software Project Management Plan (SPMP)

MAJOR SECTION HEADINGS	SECTION TOPICS
Overview	Purpose, scope, and objectives; assumptions and constraints; project deliverables; schedule and budget summary; evolution of the plan
Project Organization	External interfaces; internal structure; roles and responsibilities
Managerial Process Plan	Start-up plans (estimation, staffing, resource acquisition, and project staff training plans); work plan (work activities, schedule, resource, and budget allocation); control plan; risk management plan; closeout plan
Technical Process Plans	Process model; methods, tools, and techniques; infrastructure plan; product acceptance plan
Supporting Process Plans	Configuration management plan; verification and validation plan; documentation plan; quality assurance plan; reviews and audits; problem resolution plan; subcontractor management plan; process improvement plan

IEEE Standard 1058-1998.

Successful Planning

"The winners clearly spell out what needs to be done in a project, by whom, when, and how. For this they use an integrated toolbox, including PM tools, methods, and techniques...If a scheduling template is developed and used over and over, it becomes a repeatable action that leads to higher productivity and lower uncertainty. Sure, using scheduling templates is neither a breakthrough nor a feat. But laggards exhibited almost no use of the templates. Rather, in constructing schedules their project managers started with a clean sheet, a clear waste of time.“*

*Milosevic, Dragan and And Ozbay. “Delivering Projects: What the Winners Do.” Proceedings of the Project Management Institute Annual Seminars & Symposium (November 2001).

Executing Process Group:

Directing and Managing Project Work

- Involves managing and performing the work described in the project management plan
- Project planning and execution are intertwined and inseparable activities

Providing Leadership and a Supportive Culture

- Project managers must lead by example to demonstrate the importance of creating and then following good project plans

Capitalizing on Product, Business, and Application Area Knowledge

- It is often helpful for IT project managers to have prior technical experience
- Small vs. Large Projects

Project Execution Tools and Techniques

- **Expert judgment**
- **Meetings**
- **Project management information systems**

Monitoring and Controlling Project Work

- Changes are inevitable on most projects, so it's important to develop and follow a process to monitor and control changes
- Monitoring project work includes collecting, measuring, and disseminating performance information
- A **baseline** is the approved project management plan plus approved changes

Performing Integrated Change Control

- Three main objectives are:
 - Influencing the factors that create changes to ensure that changes are beneficial
 - Determining that a change has occurred
 - Managing actual changes as they occur

Change Control on Information Technology Projects

- Former view & Problems
- Modern view & Solution

Change Control System

- A **change control system** is a formal, documented process that describes when and how official project documents and work may be changed
- Describes who is authorized to make changes and how to make them

Change Control Board (CCB)

- A **change control board** is a formal group of people responsible for approving or rejecting changes on a project
- CCBs provide guidelines for preparing change requests, evaluate change requests, and manage the implementation of approved changes

Configuration Management

- **Configuration management** ensures that the descriptions of the project's products are correct and complete
- Involves identifying and controlling the functional and physical design characteristics of products and their support documentation

Table 4-3. Suggestions for Performing Integrated Change Control

View project management as a process of constant communication and negotiation.

Plan for change.

Establish a formal change control system, including a change control board (CCB).

Use effective configuration management.

Define procedures for making timely decisions on smaller changes.

Use written and oral performance reports to help identify and manage change.

Use project management and other software to help manage and communicate changes.

Focus on leading the project team and meeting overall project goals and expectations.

Closing Projects or Phases

- To close a project or phase, you must finalize all activities and transfer the completed or cancelled work to the appropriate people
- Main outputs include
 - Final product, service, or result transition
 - Organizational process asset updates

Using Software to Assist in Project Integration Management

- Several types of software can be used to assist in project integration management
 - Project management software can pull everything together and show detailed and summarized information
 - **Business Service Management (BSM)** tools track the execution of business process flows

Chapter Summary

- Project integration management involves coordinating all of the other knowledge areas throughout a project's life cycle
- Main processes include
 - Develop the project charter
 - Develop the project management plan
 - Direct and manage project execution
 - Monitor and control project work
 - Perform integrated change control
 - Close the project or phase

Sample CAPM Questions

Which of the following issues the project charter document?

- a. The performing organization's higher management
- b. Any stakeholder
- c. The customer
- d. The project manager

What document is the result of the project initiation group?

- a. Statement of Work
- b. Project Charter
- c. Scope Plan Document
- d. Preliminary Scope Document

Sample CAPM Questions

The integrated change control process is used to manage changes to the project at which stage?

- a. From initiating through closing
- b. Planning only
- c. Executing only
- d. Executing and closing only

Which of the following stakeholders can authorize closure?

- a. Project Manager
- b. Customer
- c. Project Sponsor
- d. Functional Manager